Uncertainty and Model Checking for Atmospheric Inversions

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Model

What we know, and what we want to know:

$$\begin{aligned} \mathbf{y} &: \text{Observed measurements} & \text{dimension } M \\ \mathbf{x} &: \text{Unobserved flux} & \text{dimension } N \end{aligned}$$

Hyperparemters and data generating process:

$$\begin{aligned} \mathbf{H} &: \text{Footprints} & \text{dimension } M \times N \\ \mathbf{B} &: \text{Prior covariance} & \text{dimension } N \times N \\ \mathbf{R} &: \text{Observation error} & \text{dimension } M \times M \\ \mathbf{m} &: \text{Prior inventory} & \text{dimension } N \end{aligned}$$

$$\begin{aligned} \mathbf{y} &\sim \mathcal{N}(\mathbf{m}, \mathbf{B}) & \text{Actual fluxes are smooth, centered at inventory} \\ \mathbf{x}|\mathbf{y} &\sim \mathcal{N}(\mathbf{Hy}, \mathbf{R}) & \text{The footprints transport } \mathbf{y}\text{, then observe with some error} \end{aligned}$$

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Inversions estimate $\hat{\mathbf{x}} = \mathbb{E}_{\mathcal{P}(\mathbf{x}|\mathbf{y})}[\mathbf{x}]$ at great expense!

Would also like, without a ton of extra computation:

- Measurements of posterior uncertainty. (E.g., how large is $\mathrm{Cov}_{\mathcal{P}(\mathbf{x}|\mathbf{y})}(\mathbf{x})$?)
- Measurements of prior sensitivity (E.g., what if the length scale of $\mathbf{B}$ changed?)
- Measurements of data influence (E.g., how much is my data, and how much my prior?)

Inference

Data generating process:

$$\mathbf{y} \sim \mathcal{N}\left(\mathbf{m}, \mathbf{B}\right) \qquad \text{Actual fluxes are smooth, centered at inventory}$$

$$\mathbf{x}|\mathbf{y} \sim \mathcal{N}\left(\mathbf{H}\mathbf{y}, \mathbf{R}\right) \qquad \text{The footprints transport } \mathbf{y}\text{, observed with some error}$$

Inference: $\mathcal{P}(\mathbf{y}, \mathbf{x})$ is Gaussian $\Rightarrow \mathcal{P}(\mathbf{x}|\mathbf{y})$ is Gaussian.

$$\mathcal{P}(\mathbf{x}|\mathbf{y}) = \mathcal{N}\left(\hat{\mathbf{x}}, \hat{\mathbf{V}}\right) \quad \text{where}$$

$$\hat{\mathbf{x}} = \mathbf{m} + \left(\mathbf{H}\mathbf{B}\right)^{\intercal}\left(\mathbf{H}\mathbf{B}\mathbf{H}^{\intercal} + \mathbf{R}\right)^{-1}\left(\mathbf{y} - \mathbf{m}\right)$$

$$\hat{\mathbf{V}} = \mathbf{B} - \left(\mathbf{H}\mathbf{B}\right)^{\intercal}\left(\mathbf{H}\mathbf{B}\mathbf{H}^{\intercal} + \mathbf{R}\right)^{-1}\left(\mathbf{H}\mathbf{B}\right)$$

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$$\hat{\mathbf{x}} = \mathbf{m} + (\mathbf{HB})^\intercal \left(\mathbf{HBH}^\intercal + \mathbf{R}\right)^{-1} (\mathbf{y} - \mathbf{m})$$

$$\hat{\mathbf{V}} = \mathbf{B} - (\mathbf{HB})^\intercal \left(\mathbf{HBH}^\intercal + \mathbf{R}\right)^{-1} (\mathbf{HB})$$

Inversions estimate $\hat{\mathbf{x}} = \mathbb{E}_{\mathcal{P}(\mathbf{x}|\mathbf{y})}[\mathbf{x}]$. This is slow because of

- Forming $\mathbf{H}$ and $\mathbf{B}$ in memory
- Computing $\mathbf{HB}$ and $\mathbf{HBH}^\intercal$
- Inverting $(\mathbf{HBH}^\intercal + \mathbf{R})$

Sequoia has sped these up, and also saved to disk what she can.

Assume we can re-use these computations. Let's get posterior uncertainty, prior sensitivity, and data influence.

Posterior uncertainty

$$\mathcal{P}(\mathbf{x}|\mathbf{y}) = \mathcal{N}\left(\hat{\mathbf{x}}, \hat{\mathbf{V}}\right) \text{ where } \hat{\mathbf{V}} \text{ is huge } (N \times N)$$

Problem: Obviously we can't compute $\hat{\mathbf{V}}$, it is huge.

Idea: We actually want to know is not $\mathbf{x}$, but linear combinations of $\mathbf{x}$.

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Example: Want s, the total emissions in hour t, summed over all spatial locations.

Let **a** be an $N-$vector with:

$$\mathbf{a}_n = \begin{cases} 1 & \text{If observation } n \text{ was at time } t \\ 0 & \text{otherwise} \end{cases}$$

Then

$$s = \sum_{n=1}^{N} \mathbf{a}_n \mathbf{x}_n = \mathbf{a}^\intercal \mathbf{x} \Rightarrow$$

$$\mathrm{Cov}_{\mathcal{P}(\mathbf{x}|\mathbf{y})}(s) = \mathrm{Cov}_{\mathcal{P}(\mathbf{x}|\mathbf{y})}(\mathbf{a}^\intercal \mathbf{x}) = \mathbf{a}^\intercal \mathrm{Cov}_{\mathcal{P}(\mathbf{x}|\mathbf{y})}(\mathbf{x})\ \mathbf{a}$$

The covariance of the scalar s is a linear combination of stuff we computed for $\hat{\mathbf{x}}$:

$$\mathrm{Cov}_{\mathcal{P}(\mathbf{x}|\mathbf{y})}(s) = \mathbf{a}^\intercal \underbrace{\mathbf{B}}_{\text{done}} \mathbf{a} - (\underbrace{\mathbf{HB}}_{\text{done}}\mathbf{a})^\intercal \underbrace{(\mathbf{HBH}^\intercal + \mathbf{R})^{-1}}_{\text{done}} (\underbrace{\mathbf{HB}}_{\text{done}}\mathbf{a})$$

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Example: Want $\mathbf{s}$, the total emissions in hours $t = 1, \ldots, 24$, summed over all spatial locations.

Let $\mathbf{A}$ be an $24 \times N$ matrix with:

$$\mathbf{A}_{tn} = \begin{cases} 1 & \text{If observation } n \text{ was at time } t \\ 0 & \text{otherwise} \end{cases}$$

Then

$$\mathbf{s} = \mathbf{A}\mathbf{x} \Rightarrow$$

$$\mathrm{Cov}_{\mathcal{P}(\mathbf{x}|\mathbf{y})}\left(\mathbf{s}\right) = \mathbf{A}\mathrm{Cov}_{\mathcal{P}(\mathbf{x}|\mathbf{y})}\left(\mathbf{x}\right)\mathbf{A}^\intercal$$

The covariance of the vector s is a linear combination of stuff we already computed.

Posterior uncertainty

$$\mathcal{P}(\mathbf{x}|\mathbf{y}) = \mathcal{N}\left(\hat{\mathbf{x}}, \hat{\mathbf{V}}\right) \text{ where } \hat{\mathbf{V}} \text{ is huge } (N \times N)$$

Problem: Obviously we can't compute $\hat{\mathbf{V}}$, it is huge.

Idea: We actually want to know is not $\mathbf{x}$, but linear combinations of $\mathbf{x}$.

Extreme example: Want t^*, the hour of the day where the total flux is largest.

The value t^* is not a linear combination of $\mathbf{x}$. But we just showed that

$$\mathcal{P}(\mathbf{s}|\mathbf{y}) = \mathcal{N}\left(\mathbf{A}\hat{\mathbf{x}}, \mathbf{A}\hat{\mathbf{V}}\mathbf{A}^\mathsf{T}\right),$$

and everything is computable. So we can:

- Draw $\mathbf{s}_1, \ldots, \mathbf{s}_K$ from the multivariate normal
- Compute $t_k^* = \text{argmax}_t\, \mathbf{s}_{kt}$
- Estimate $\text{Var}_{\mathcal{P}(\mathbf{x}|\mathbf{y})}\left(t^*\right) \approx \frac{1}{K}\sum_{k=1}^{K}(t_k^* - \overline{t^*})^2$

The covariance of the vector t^* easily estimated from stuff we already computed.

Posterior Variance Results

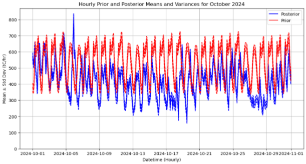

Figure 11: Hourly posterior mean and variance for a month of inversions

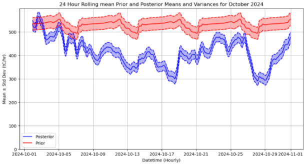

Figure 12: Smoothed hourly posterior mean and variance for a month of inversions

Prior sensitivity

The prior $\mathbf{B}$ depends on quantities like ρ, an assumed lengthscale. So we might write

$$\hat{\mathbf{x}}(\rho) = \mathbf{m} + \left(\mathbf{H}\mathbf{B}(\rho)\right)^\intercal \left(\mathbf{H}\mathbf{B}(\rho)\mathbf{H}^\intercal + \mathbf{R}\right)^{-1} (\mathbf{y} - \mathbf{m})$$

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Problem:

- Re-computing $\hat{\mathbf{x}}(\rho)$ for a new lengthscale is expensive (a whole new problem).
- There actually can be lots of hyperparameters, and you don't know which matter.

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- Re-computing $\hat{\mathbf{x}}(\rho)$ for a new lengthscale is expensive (a whole new problem).
- There actually can be lots of hyperparameters, and you don't know which matter.

Idea: Starting from a solution at the initial guess ρ_0, form a Taylor series expansion

$$\hat{\mathbf{x}}(\rho) \approx \hat{\mathbf{x}} + \frac{\partial \hat{\mathbf{x}}(\rho)}{\partial \rho} (\rho - \rho_0)$$

$$\frac{\partial}{\partial \rho} \left(\mathbf{H}\mathbf{B}(\rho)\mathbf{H}^\intercal + \mathbf{R}\right)^{-1} = -\underbrace{\left(\mathbf{H}\mathbf{B}\mathbf{H}^\intercal + \mathbf{R}\right)^{-1}}_{\text{done}} \mathbf{H} \frac{\partial \mathbf{B}(\rho)}{\partial \rho} \mathbf{H}^\intercal \underbrace{\left(\mathbf{H}\mathbf{B}\mathbf{H}^\intercal + \mathbf{R}\right)^{-1}}_{\text{done}}$$

The Taylor series can be easily estimated from stuff we already computed.

Prior sensitivity

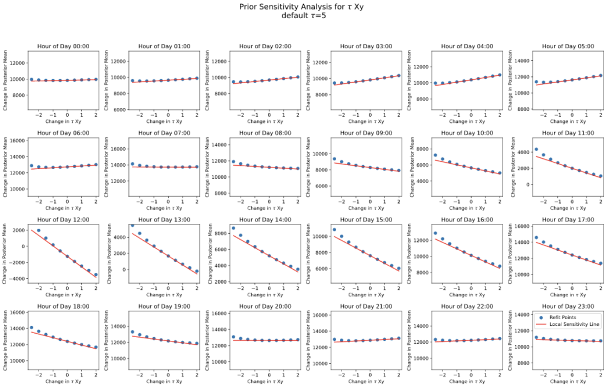

Figure 20: Prior sensitivity with respect to $\boldsymbol{\tau}_{xy}$ for a single inversion

Data influence

Problem: In inversion problems, the prior is informative, and the model does smoothing. How is your data being used? What patterns are possible to identify?

Example question: Can I detect emissions from road pixels?

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Example question: Can I detect emissions from road pixels?

Idea: Let $\mathbf{r}$ be an N-vector encoding road pixels.

$$\mathbf{r}_n = \begin{cases} 1 & \text{If observation } n \text{ was a road pixel} \\ 0 & \text{otherwise} \end{cases}$$

Suppose we had observed not $\mathbf{y}$, but $\mathbf{y}' = \mathbf{y} + \epsilon\mathbf{r}$. That is, $\mathbf{y}'$ is a little more "roady" than $\mathbf{y}$.

Then, suppose I used $\mathbf{y}'$ to compute $\mathbf{x}'$. How much more "roady" is $\mathbf{x}'$ than $\mathbf{x}$?

$$\begin{aligned} \frac{\text{Inferred road pixel change}}{\text{Actual road pixel change}} &= \frac{\mathbf{r}^\intercal(\mathbf{x}' - \mathbf{x})}{\mathbf{r}^\intercal(\mathbf{y}' - \mathbf{y})} \\ &= \epsilon^{-1}\frac{\mathbf{r}^\intercal(\mathbf{x}' - \mathbf{x})}{\mathbf{r}^\intercal\mathbf{r}} \\ &= \frac{\mathbf{r}^\intercal \overbrace{(\mathbf{HB})^\intercal\left(\mathbf{HBH}^\intercal + \mathbf{R}\right)^{-1}}^{\text{done}} \mathbf{r}}{\mathbf{r}^\intercal\mathbf{r}} \end{aligned}$$

This is just the "gain matrix" queried for a particular output.

The data influence can be easily estimated from stuff we already computed.

Data influence

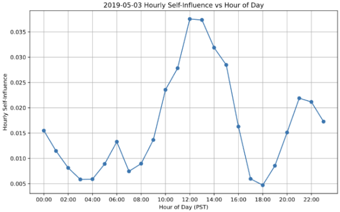

Figure 16: Hourly self-influence for a single inversion

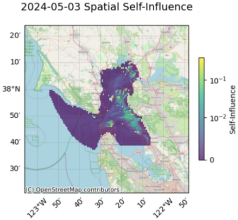

Figure 17: Spatial self-influence for a single inversion